



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 20

Subject: Programming with Java

ASSIGNMENT No : 20

TITLE :Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Problem Statement

Develop a Java application to establish connectivity with a database and perform INSERT, UPDATE, and DELETE operations.

Learning Objectives

To perform database modification operations using JDBC.

Theory

JDBC allows Java applications to perform database manipulation using SQL Data Manipulation Language (DML) statements such as INSERT, UPDATE, and DELETE. The INSERT statement adds new records into database tables, UPDATE modifies existing records, and DELETE removes unwanted records. These operations are executed using the Statement or PreparedStatement interface. PreparedStatement is generally preferred because it supports parameterized queries, improves performance, and prevents SQL injection attacks. Proper transaction management ensures that database modifications are performed safely and consistently. These operations form the basis of almost every database-driven application such as banking systems, hospital management systems, and student information systems.



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Code and output screenshots :

SQL database and table creation :

```
1 CREATE DATABASE studentdb;
2
3 • USE studentdb;
4
5 • CREATE TABLE student (
6     id INT PRIMARY KEY,
7     name VARCHAR(50),
8     marks INT
9 );
10
11 • INSERT INTO student VALUES
12 (101, 'Rahul', 85),
13 (103, 'Priya', 90),
14 (104, 'Sneha', 78);
15
16 • SELECT * FROM student;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	id	name	marks
▶	101	Rahul	85
	103	Priya	90
	104	Sneha	78
*	NULL	NULL	NULL

Result Grid
Form Editor
Field Types



Code :

```
1  import java.sql.*;
2  public class CRUDEXample {
    Run | Debug | Run main | Debug main
3      public static void main(String[] args) {
4
5          String url = "jdbc:mysql://localhost:3306/studentdb";
6          String user = "root";
7          String password = "Abcd@1234";
8
9          try {
10             Class.forName("com.mysql.cj.jdbc.Driver");
11             Connection con = DriverManager.getConnection(url, user, password);
12             Statement stmt = con.createStatement();
13
14             System.out.println("Initial table");
15             ResultSet rs = stmt.executeQuery("SELECT * FROM student");
16             while (rs.next())
17             {
18                 int id = rs.getInt("id");
19                 String name = rs.getString("name");
20                 int marks = rs.getInt("marks");
21                 System.out.println(id + " " + name + " " + marks);
22             }
23
24             // INSERT
25             String insertQuery = "INSERT INTO student VALUES (?, ?, ?)";
26             PreparedStatement ps1 = con.prepareStatement(insertQuery);
27             ps1.setInt(1, 102);
28             ps1.setString(2, "Amit");
29             ps1.setInt(3, 88);
30             ps1.executeUpdate();
```



```
31      System.out.println("Updated Table after INSERT query:");
32      rs = stmt.executeQuery("SELECT * FROM student");
33      while (rs.next())
34      {
35          int id = rs.getInt("id");
36          String name = rs.getString("name");
37          int marks = rs.getInt("marks");
38          System.out.println(id + " " + name + " " + marks);
39      }
40
41      // UPDATE
42      String updateQuery = "UPDATE student SET marks=? WHERE id=?";
43      PreparedStatement ps2 = con.prepareStatement(updateQuery);
44      ps2.setInt(1, 95);
45      ps2.setInt(2, 102);
46      ps2.executeUpdate();
47
48      System.out.println("Updated Table after UPDATE query:");
49      rs = stmt.executeQuery("SELECT * FROM student");
50      while (rs.next())
51      {
52          int id = rs.getInt("id");
53          String name = rs.getString("name");
54          int marks = rs.getInt("marks");
55          System.out.println(id + " " + name + " " + marks);
56      }
57
```



```
58 // DELETE
59 String deleteQuery = "DELETE FROM student WHERE id=?";
60 PreparedStatement ps3 = con.prepareStatement(deleteQuery);
61 ps3.setInt(1, 102);
62 ps3.executeUpdate();
63 System.out.println("Updated Table aftre DELETE query:");
64 rs = stmt.executeQuery("SELECT * FROM student");
65 while (rs.next())
66 {
67     int id = rs.getInt("id");
68     String name = rs.getString("name");
69     int marks = rs.getInt("marks");
70     System.out.println(id + " " + name + " " + marks);
71 }
72
73 con.close();
74 System.out.println("CRUD operations done!");
75
76 } catch (Exception e) {
77     System.out.println(e);
78 }
79 }
80 }
81
82
```



Output :

```
Problems 15 Output Debug Console Terminal Ports
● PS C:\Users\LENOVO\Desktop\JDBC> javac -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" CRUExample.java
● PS C:\Users\LENOVO\Desktop\JDBC> java -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" CRUExample
101 Rahul 85
103 Priya 90
104 Sneha 78
Updated Table after INSERT query:
101 Rahul 85
102 Amit 88
103 Priya 90
104 Sneha 78
Updated Table after UPDATE query:
101 Rahul 85
102 Amit 95
103 Priya 90
104 Sneha 78
Updated Table after DELETE query:
101 Rahul 85
103 Priya 90
104 Sneha 78
CRUD operations done!
● PS C:\Users\LENOVO\Desktop\JDBC> javac -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" CRUExample.java
● PS C:\Users\LENOVO\Desktop\JDBC> java -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" CRUExample
Initial table
101 Rahul 85
103 Priya 90
104 Sneha 78
Updated Table after INSERT query:
101 Rahul 85
102 Amit 88
103 Priya 90
104 Sneha 78
Updated Table after UPDATE query:
101 Rahul 85
102 Amit 95
103 Priya 90
104 Sneha 78
Updated Table after DELETE query:
101 Rahul 85
103 Priya 90
104 Sneha 78
CRUD operations done!
❖ PS C:\Users\LENOVO\Desktop\JDBC>
```

Conclusion : Thus, the experiment demonstrates database connectivity using JDBC and the use of INSERT, UPDATE, and DELETE SQL operations for managing records in a database. It provides an understanding of how database records can be added, modified, and removed through Java.



Exercise

Q1) 1. Create a Java application to manage employee records using CRUD operations

Sol) Create database :

The screenshot shows a database management interface with a SQL editor and a result grid. The SQL editor contains the following queries:

```
1 • CREATE DATABASE employeeedb;
2 • USE employeeedb;
3
4 • CREATE TABLE employee (
5     id INT PRIMARY KEY,
6     name VARCHAR(50),
7     department VARCHAR(50),
8     salary DOUBLE
9 );
10
11 • INSERT INTO employee VALUES
12 (101, 'Rahul', 'IT', 45000),
13 (102, 'Priya', 'HR', 40000),
14 (103, 'Amit', 'Sales', 35000);
15
16 • SELECT * FROM employee;
```

The result grid displays the data from the 'employee' table:

	id	name	department	salary
▶	101	Rahul	IT	45000
	102	Priya	HR	40000
	103	Amit	Sales	35000
•	NULL	NULL	NULL	NULL



Code : (Written)

```
import java.sql.*;

public class EmployeeCRUD {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/employeedb";
        String user = "root";
        String password = "root";

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, user, password);

            Statement stmt = con.createStatement();

            ResultSet rs = stmt.executeQuery("SELECT * FROM employee");

            while (rs.next()) {

                int id = rs.getInt("id");
                String name = rs.getString("name");
                String department = rs.getString("department");
                int salary = rs.getInt("salary");

                System.out.println(id + " " + name + " " + department + " " + salary);
            }

            // INSERT

            String insertQuery = "INSERT INTO employee VALUES (?, ?, ?, ?)";

            PreparedStatement ps1 = con.prepareStatement(insertQuery);

            ps1.setInt(1, 104);
```



```
ps1.setString(2, "Sneha");
ps1.setString(3, "Finance");
ps1.setInt(4, 42000);

ps1.executeUpdate();

System.out.println("Updated Table after INSERT query:");

rs = stmt.executeQuery("SELECT * FROM employee");

while (rs.next()) {

    int id = rs.getInt("id");
    String name = rs.getString("name");
    String department = rs.getString("department");
    int salary = rs.getInt("salary");

    System.out.println(id + " " + name + " " + department + " " + salary);
}

// UPDATE

String updateQuery = "UPDATE employee SET salary=? WHERE id=?";

PreparedStatement ps2 = con.prepareStatement(updateQuery);

ps2.setInt(1, 50000);
ps2.setInt(2, 104);

ps2.executeUpdate();

System.out.println("Updated Table after UPDATE query:");

rs = stmt.executeQuery("SELECT * FROM employee");

while (rs.next()) {

    int id = rs.getInt("id");
    String name = rs.getString("name");
```



```
String department = rs.getString("department");
int salary = rs.getInt("salary");

System.out.println(id + " " + name + " " + department + " " + salary);
}

// DELETE

String deleteQuery = "DELETE FROM employee WHERE id=?";

PreparedStatement ps3 = con.prepareStatement(deleteQuery);

ps3.setInt(1, 104);

ps3.executeUpdate();

System.out.println("Updated Table after DELETE query:");

rs = stmt.executeQuery("SELECT * FROM employee");

while (rs.next()) {

    int id = rs.getInt("id");
    String name = rs.getString("name");
    String department = rs.getString("department");
    int salary = rs.getInt("salary");

    System.out.println(id + " " + name + " " + department + " " + salary);
}

con.close();

System.out.println("CRUD operations done!");

} catch (Exception e) {

    System.out.println(e);

}
```



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} }

Output :

```
Problems 6 Output Debug Console Terminal Ports
• PS C:\Users\LENOVO\Desktop\JDBC> javac -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" EmployeeCRUD.java
• PS C:\Users\LENOVO\Desktop\JDBC> java -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" EmployeeCRUD
101 Rahul IT 45000
102 Priya HR 40000
103 Amit Sales 35000
Updated Table after INSERT query:
101 Rahul IT 45000
102 Priya HR 40000
103 Amit Sales 35000
104 Sneha Finance 42000
Updated Table after UPDATE query:
101 Rahul IT 45000
102 Priya HR 40000
103 Amit Sales 35000
104 Sneha Finance 50000
Updated Table after DELETE query:
101 Rahul IT 45000
102 Priya HR 40000
103 Amit Sales 35000
CRUD operations done!
PS C:\Users\LENOVO\Desktop\JDBC>
```



Q2) Create a Java application to manage student records using CRUD operations. Perform Create, Read, Update, and Delete operations on student details such as roll number, name, course, and marks using a database.

Sol) Create Database :

The screenshot shows a database management interface with a SQL editor and a result grid. The SQL editor contains the following queries:

```
1 • CREATE DATABASE studentdb;
2
3 • USE studentdb;
4
5 • CREATE TABLE student (
6     roll_no INT PRIMARY KEY,
7     name VARCHAR(50),
8     course VARCHAR(50),
9     marks INT
10 );
11
12 • INSERT INTO student VALUES
13     (1, 'Rahul', 'BCA', 85),
14     (2, 'Priya', 'BCA', 90),
15     (3, 'Amit', 'BSc CS', 78);
16
17 • SELECT * FROM student;
```

The result grid displays the data inserted into the 'student' table:

roll_no	name	course	marks
1	Rahul	BCA	85
2	Priya	BCA	90
3	Amit	BSc CS	78
NULL	NULL	NULL	NULL



Code (Written) :

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.Statement;

public class StudentCRUD {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/studentdb";
        String user = "root";
        String password = "Abcd@1234";

        try {

            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(url, user, password);

            Statement stmt = con.createStatement();

            // DISPLAY EXISTING RECORDS

            ResultSet rs = stmt.executeQuery("SELECT * FROM student");

            while (rs.next()) {

                int roll = rs.getInt("roll_no");
                String name = rs.getString("name");
                String course = rs.getString("course");
                int marks = rs.getInt("marks");

                System.out.println(roll + " " + name + " " + course + " " + marks);
            }

            // INSERT
```



```
String insertQuery = "INSERT INTO student VALUES (?, ?, ?, ?)";
```

```
PreparedStatement ps1 = con.prepareStatement(insertQuery);
```

```
ps1.setInt(1, 4);  
ps1.setString(2, "Sneha");  
ps1.setString(3, "BCA");  
ps1.setInt(4, 88);
```

```
ps1.executeUpdate();
```

```
System.out.println("Updated Table after INSERT query:");
```

```
rs = stmt.executeQuery("SELECT * FROM student");
```

```
while (rs.next()) {
```

```
    int roll = rs.getInt("roll_no");  
    String name = rs.getString("name");  
    String course = rs.getString("course");  
    int marks = rs.getInt("marks");
```

```
    System.out.println(roll + " " + name + " " + course + " " + marks);  
}
```

```
// UPDATE
```

```
String updateQuery = "UPDATE student SET marks=? WHERE roll_no=?";
```

```
PreparedStatement ps2 = con.prepareStatement(updateQuery);
```

```
ps2.setInt(1, 95);  
ps2.setInt(2, 4);
```

```
ps2.executeUpdate();
```

```
System.out.println("Updated Table after UPDATE query:");
```



```
rs = stmt.executeQuery("SELECT * FROM student");

while (rs.next()) {

    int roll = rs.getInt("roll_no");
    String name = rs.getString("name");
    String course = rs.getString("course");
    int marks = rs.getInt("marks");

    System.out.println(roll + " " + name + " " + course + " " + marks);
}

// DELETE

String deleteQuery = "DELETE FROM student WHERE roll_no=?";

PreparedStatement ps3 = con.prepareStatement(deleteQuery);

ps3.setInt(1, 4);

ps3.executeUpdate();

System.out.println("Updated Table after DELETE query:");

rs = stmt.executeQuery("SELECT * FROM student");

while (rs.next()) {

    int roll = rs.getInt("roll_no");
    String name = rs.getString("name");
    String course = rs.getString("course");
    int marks = rs.getInt("marks");

    System.out.println(roll + " " + name + " " + course + " " + marks);
}

con.close();

System.out.println("CRUD operations done!");
```



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```
    } catch (Exception e) {  
  
        System.out.println(e);  
  
    }  
}  
}
```

Output :

```
Problems 9 Output Debug Console Terminal Ports  
● PS C:\Users\LENOVO\Desktop\JDBC> javac -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" StudentCRUD.java  
● PS C:\Users\LENOVO\Desktop\JDBC> java -cp ".;mysql-connector-j-26.7.0\mysql-connector-j-26.7.0\mysql-connector-j-26.7.0.jar" StudentCRUD  
1 Rahul BCA 85  
2 Priya BCA 90  
3 Amit BSc CS 78  
Updated Table after INSERT query:  
1 Rahul BCA 85  
2 Priya BCA 90  
3 Amit BSc CS 78  
4 Sneha BCA 88  
Updated Table after UPDATE query:  
1 Rahul BCA 85  
2 Priya BCA 90  
3 Amit BSc CS 78  
4 Sneha BCA 95  
Updated Table after DELETE query:  
1 Rahul BCA 85  
2 Priya BCA 90  
3 Amit BSc CS 78  
CRUD operations done!  
❖ PS C:\Users\LENOVO\Desktop\JDBC>
```